advanced qm notes

beann

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1 Introduction

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Lecture 1: Review of Quantum Mechanics

Symmetries, the Schrödinger Equation, and Generators Based on Leonard Susskind's *Advanced Quantum Mechanics*

2 1. Review of the Schrödinger Equation

The time-dependent Schrödinger equation governs how the state vector $\psi(t)$ evolves:

$$i\hbar \frac{\partial}{\partial t}\psi(t) = H\psi(t),$$

where H is the Hamiltonian operator. The formal solution is

$$\psi(t) = e^{-iHt/\hbar}\psi(0).$$

The operator $U(t) = e^{-iHt/\hbar}$ is unitary, ensuring that the norm $\psi\psi = 1$ is preserved at all times.

2.1 1.1 Time-independent Schrödinger equation

If we expand ψ in energy eigenstates $HE_n = E_nE_n$, then the time evolution becomes:

$$\psi(t) = \sum_n c_n\, e^{-iE_nt/\hbar}E_n,$$

where $c_n = E_n\psi(0)$. Each component oscillates at its own frequency $\omega_n = E_n/\hbar$.

2.2 1.2 Expectation values

For any observable L, its expectation value evolves as:

$$\frac{d}{dt}L = \frac{i}{\hbar}[H, L].$$

If $[H, L] = 0$, then L is a constant of the motion—the observable is *conserved*.

3 2. Symmetry in Quantum Mechanics

Definition 1 (Symmetry) *A* ***symmetry*** *of a quantum system is a unitary operator U that commutes with the Hamiltonian:*

$$[U, H] = 0.$$

Equivalently, if $\psi(t)$ is a solution of the Schrödinger equation, then $U\psi(t)$ is also a solution.

Symmetries come in two varieties:

[nosep]

- **Discrete symmetries**: reflections, parity, permutations. The operator U acts by a finite transformation.
- **Continuous symmetries**: translations, rotations. The operator $U(\epsilon)$ depends on a continuous parameter and can be built up from infinitesimal steps.

4 3. Continuous Symmetries and Generators

A continuous symmetry is generated by a Hermitian operator. For an infinitesimal transformation parameterized by ϵ:

$$U(\epsilon) = I - i\epsilon G,$$

where $G = G^\dagger$ is the **generator** of the symmetry. The condition $[U, H] = 0$ implies $[G, H] = 0$, so the generator is a conserved quantity.

4.1 3.1 Translation symmetry and momentum

Translation by a distance δ acts on wave functions as:

$$T(\delta)\psi(x) = \psi(x - \delta).$$

Expanding to first order:

$$\psi(x - \delta) \approx \psi(x) - \delta\frac{\partial\psi}{\partial x} = \left(1 - \delta\frac{\partial}{\partial x}\right)\psi(x).$$

Comparing with $U(\delta) = I - i\delta G/\hbar$, we identify the generator:

$$G = -i\hbar \frac{\partial}{\partial x} = p,$$

which is the **momentum operator**. If the Hamiltonian is translationally invariant ($[T(\delta), H] = 0$), then momentum is conserved.

4.2 3.2 Noether's theorem (quantum version)

The connection between symmetries and conservation laws:

[nosep]

1. Every continuous symmetry has an associated Hermitian generator G.
2. $[U, H] = 0 \iff [G, H] = 0$.
3. $[G, H] = 0$ means G is conserved: $\frac{d}{dt}G = 0$.

This is the quantum-mechanical version of Noether's theorem: **every continuous symmetry implies a conservation law**, and vice versa.

5 4. Finite Transformations from Infinitesimal Ones

A finite translation by a is built by iterating infinitesimal translations:

$$T(a) = \lim_{N\to\infty} \left(1 - \frac{ia}{\hbar N} p\right)^N = e^{-iap/\hbar}.$$

More generally, for a symmetry generated by G with parameter θ:

$$U(\theta) = e^{-i\theta G/\hbar}.$$

This is a unitary operator because G is Hermitian.

6 5. Summary of Key Correspondences

Symmetry	**Generator**	**Conserved Quantity**
Time translation	Hamiltonian H	Energy
Space translation	Momentum p	Momentum
Rotation	Angular momentum L	Angular momentum

The unifying principle: the Hamiltonian generates time evolution, and any operator that commutes with it is conserved. The structure of symmetries and their generators is the foundation on which the rest of the course is built.

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Lecture 2: Symmetry, Degeneracy, and Angular Momentum

Rotation Generators and the Angular Momentum Algebra Based on Leonard Susskind's *Advanced Quantum Mechanics*

7 1. Symmetry and Degeneracy

When a Hamiltonian possesses a symmetry $[U, H] = 0$, the energy eigenstates come in *multiplets*—sets of states related by the symmetry operator that all share the same energy.

Definition 2 (Degeneracy) *An energy level E_n is **degenerate** if there exist multiple linearly independent eigenstates with the same eigenvalue E_n. The number of such states is the **degree of degeneracy**.*

If $HE = EE$ and $[U, H] = 0$, then:

$$H\big(UE\big) = U\big(HE\big) = E\big(UE\big).$$

So UE is also an eigenstate with the same energy E. Unless UE is proportional to E, the level is degenerate.

Example: For a rotationally invariant Hamiltonian, all states related by rotations share the same energy. This leads to the characteristic $(2\ell + 1)$-fold degeneracy of angular momentum multiplets.

8 2. Angular Momentum as a Generator of Rotations

8.1 2.1 Classical angular momentum

In classical mechanics, the angular momentum of a particle with position r and momentum p is:

$$L = r \times p.$$

In component form:

$$L_x = yp_z - zp_y, \qquad L_y = zp_x - xp_z, \qquad L_z = xp_y - yp_x.$$

Angular momentum is the generator of rotations: a rotation by angle θ about axis $\hat{n}$ is generated by $L \cdot \hat{n}$.

8.2 2.2 Quantum angular momentum operators

Promoting to operators via $p_i \to -i\hbar\,\partial/\partial x_i$:

$$\hat{L}_x = -i\hbar\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right), \quad \hat{L}_y = -i\hbar\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}\right), \quad \hat{L}_z = -i\hbar\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right).$$

A finite rotation by angle θ about the z-axis is:

$$R_z(\theta) = e^{-i\theta L_z/\hbar}.$$

9 3. The Angular Momentum Commutation Relations

The fundamental commutation relations of angular momentum are:

$$[L_x, L_y] = i\hbar L_z, \qquad [L_y, L_z] = i\hbar L_x, \qquad [L_z, L_x] = i\hbar L_y.$$

Compactly written using the Levi-Civita symbol:

$$[L_i, L_j] = i\hbar\,\epsilon_{ijk}\,L_k.$$

These commutation relations encode the geometry of rotations: rotations about different axes do not commute, and the commutator of two rotation generators yields the third.

9.1 3.1 The total angular momentum operator

Define the operator:

$$L^2 = L_x^2 + L_y^2 + L_z^2.$$

A crucial property is that L^2 commutes with every component:

$$[L^2, L_i] = 0 \quad for i = x, y, z.$$

Therefore, we can simultaneously diagonalize L^2 and *one* component of L (conventionally L_z). However, the individual components do not commute with each other, so we cannot simultaneously know all three.

10 4. Raising and Lowering Operators

Definition 3 (Ladder operators) *The angular momentum* ***raising*** *and* ***lowering*** *operators are:*

$$L_+ = L_x + iL_y, \qquad L_- = L_x - iL_y.$$

Note that $L_+^\dagger = L_-$.

10.1 4.1 Key commutation relations

$[\mathrm{L}_z, L_+] = +\hbar\, L_+,$
$\quad \mathrm{L}_z, L_-$
$= -\hbar\, L_-,$
$\quad \mathrm{L}_+, L_-$
$= 2\hbar\, L_z.$

10.2 4.2 Action on eigenstates

If $L_z m = m\hbar m$, then:

$$L_z\big(L_+ m\big) = (m+1)\hbar\big(L_+ m\big), \qquad L_z\big(L_- m\big) = (m-1)\hbar\big(L_- m\big).$$

So L_+ raises the L_z eigenvalue by $\hbar$, and L_- lowers it by $\hbar$. This is the "ladder" that generates the full spectrum of L_z eigenvalues within a multiplet.

11 5. The Spectrum of Angular Momentum

11.1 5.1 Bounded spectrum

Since $L^2 = L_z^2 + 12(L_+L_- + L_-L_+)$ and all terms are non-negative in expectation, the eigenvalues of L_z must be bounded:

$$-\ell \le m \le \ell,$$

where ℓ is determined by $L^2\ell, m = \hbar^2\ell(\ell+1)\ell, m$.

11.2 5.2 Integer and half-integer values

The ladder terminates: there must exist a top state ℓ, ℓ with $L_+\ell, \ell = 0$ and a bottom state $\ell, -\ell$ with $L_-\ell, -\ell = 0$. The requirement that the ladder has an integer number of steps forces:

$$\ell = 0,\ 12,\ 1,\ 32,\ 2,\ \ldots$$

For each ℓ, the magnetic quantum number m takes the $2\ell + 1$ values:

$$m = -\ell,\ -\ell+1,\ \ldots,\ \ell-1,\ \ell.$$

11.3 5.3 Degeneracy from rotational symmetry

For a rotationally invariant Hamiltonian ($[H, L] = 0$), all $2\ell + 1$ states in a multiplet have the same energy. The energy depends on ℓ but not on m.

The algebraic structure developed here—commutation relations, ladder operators, bounded spectra—is one of the most powerful tools in quantum mechanics. It will be applied to the hydrogen atom and the harmonic oscillator in the next lecture.

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Lecture 3: Central Forces and the Hydrogen Atom

Angular Momentum Multiplets, the Coulomb Potential, and the Harmonic Oscillator Based on Leonard Susskind's *Advanced Quantum Mechanics*

12 1. Angular Momentum Multiplets

Recall that the simultaneous eigenstates of L^2 and L_z are labeled ℓ, m with:

$$L^2 \ell, m = \hbar^2 \ell(\ell+1)\ell, m, \qquad L_z \ell, m = m\hbar \ell, m.$$

For each value of ℓ, there are $2\ell + 1$ states with $m = -\ell, -\ell+1, \ldots, \ell$. These form a **multiplet**—a complete set of states related by the raising and lowering operators $L_\pm$.

In a rotationally invariant system, the energy depends on ℓ but not on m, giving a $(2\ell+1)$-fold degeneracy for each angular momentum quantum number.

13 2. The Central Force Problem

A central force is one in which the potential depends only on the distance from the origin: $V(r) = V(r)$. The Hamiltonian is:

$$H = \frac{p^2}{2m} + V(r) = -\frac{\hbar^2}{2m}\nabla^2 + V(r).$$

Because V depends only on r, the system is rotationally invariant: $[H, L] = 0$.

13.1 2.1 Separation of variables

In spherical coordinates, the Laplacian separates into radial and angular parts:

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial}{\partial r}\right) - \frac{L^2}{\hbar^2 r^2}.$$

The wave function factorizes as $\psi(r, \theta, \phi) = R(r)\, Y_\ell^m(\theta, \phi)$, where Y_ℓ^m are the spherical harmonics—the eigenfunctions of L^2 and L_z.

13.2 2.2 The radial equation

Substituting the separation into the Schrödinger equation yields the radial equation:

$$-\frac{\hbar^2}{2m}\left[\frac{d^2 u}{dr^2} - \frac{\ell(\ell+1)}{r^2} u\right] + V(r)\, u = E\, u,$$

where $u(r) = rR(r)$. The term $\hbar^2 \ell(\ell+1)/(2mr^2)$ acts as a **centrifugal barrier** that pushes particles with higher angular momentum away from the origin.

14 3. The Hydrogen Atom

For the hydrogen atom, the Coulomb potential is:

$$V(r) = -\frac{e^2}{r}.$$

14.1 3.1 Energy levels

The bound-state energies are:

$$E_n = -\frac{me^4}{2\hbar^2}\frac{1}{n^2} = -\frac{13.6\ eV}{n^2},$$

where $n = 1, 2, 3, \ldots$ is the **principal quantum number**.

14.2 3.2 Nodes and quantum numbers

The radial wave function $R_{n\ell}(r)$ has $n - \ell - 1$ nodes (zeros). The number of nodes determines how rapidly the wave function oscillates, which is directly connected to the kinetic energy.

For a given n, the allowed values of ℓ are:

$$\ell = 0, 1, 2, \ldots, n - 1.$$

14.3 3.3 Special degeneracy of the Coulomb potential

For a general central potential, the energy depends on both n and ℓ. The Coulomb potential is special: the energy depends *only* on n. States with different ℓ but the same n are degenerate.

The total degeneracy for principal quantum number n is:

$$\sum_{\ell=0}^{n-1}(2\ell + 1) = n^2.$$

This "accidental" degeneracy arises from a hidden symmetry of the $1/r$ potential (the Laplace–Runge–Lenz vector), beyond ordinary rotational symmetry.

15 4. Introduction to the Harmonic Oscillator

The harmonic oscillator is one of the most important systems in all of physics. It appears in:

[nosep]

- Vibrations of molecules and crystals (phonons).
- Modes of the electromagnetic field (photons).
- Small oscillations about any stable equilibrium.
- The foundation of quantum field theory.

15.1 4.1 Classical harmonic oscillator

The classical Hamiltonian is:

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2,$$

where ω is the angular frequency. The classical solutions are sinusoidal oscillations at frequency ω.

15.2 4.2 Quantum treatment preview

The quantum treatment introduces **creation** and **annihilation** operators (also called ladder operators) that algebraically generate the entire energy spectrum:

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right), \qquad n = 0, 1, 2, \ldots$$

The energy levels are equally spaced with separation $\hbar\omega$, and the ground-state energy $E_0 = 12\hbar\omega$ (the **zero-point energy**) is nonzero.

The algebraic approach to the harmonic oscillator—using creation and annihilation operators—will be developed fully in the next lecture. This algebraic machinery will later become the foundation of quantum field theory.

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Lecture 4: The Harmonic Oscillator and Spin

Wave Functions, Creation/Annihilation Operators, and Spin-$\frac{1}{2}$ Based on Leonard Susskind's *Advanced Quantum Mechanics*

16 1. Algebraic Structure of the Harmonic Oscillator

The harmonic oscillator Hamiltonian is:

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2.$$

The entire spectrum can be derived algebraically using the creation and annihilation operators.

Definition 4 (Creation and annihilation operators) *Define:*

$$a = \sqrt{\frac{m\omega}{2\hbar}}\left(x + \frac{ip}{m\omega}\right), \qquad a^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(x - \frac{ip}{m\omega}\right).$$

These satisfy the canonical commutation relation:

$$[a, a^\dagger] = 1.$$

16.1 1.1 The number operator

The **number operator** is:

$$N = a^\dagger a, \qquad Nn = nn.$$

The Hamiltonian expressed in terms of N:

$$H = \hbar\omega\left(N + \frac{1}{2}\right) = \hbar\omega\left(a^\dagger a + \frac{1}{2}\right).$$

16.2 1.2 Commutation relations with N

[N, a] = -a,
N, a†
= +a†. These show that a lowers the eigenvalue of N by one, and $a^\dagger$ raises it by one:

$$a^\dagger n = \sqrt{n+1}\, n + 1, \qquad an = \sqrt{n}\, n - 1.$$

16.3 1.3 The ground state

The ground state 0 is defined by $a0 = 0$. All excited states are built by repeated application of $a^\dagger$:

$$n = \frac{(a^\dagger)^n}{\sqrt{n!}} 0.$$

17 2. Harmonic Oscillator Wave Functions

17.1 2.1 Ground-state wave function

The condition $a0 = 0$ translates in position space to:

$$\left(x + \frac{\hbar}{m\omega}\frac{d}{dx}\right)\psi_0(x) = 0.$$

The solution is a Gaussian:

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{m\omega x^2}{2\hbar}\right).$$

17.2 2.2 Excited-state wave functions

Higher wave functions are obtained by applying the creation operator:

$$\psi_n(x) = \frac{1}{\sqrt{n!}} \left(\frac{m\omega}{2\hbar}\right)^{n/2} \left(x - \frac{\hbar}{m\omega}\frac{d}{dx}\right)^n \psi_0(x).$$

These are Gaussian functions multiplied by Hermite polynomials H_n:

$$\psi_n(x) \propto H_n\left(\sqrt{\frac{m\omega}{\hbar}}\, x\right) \exp\left(-\frac{m\omega x^2}{2\hbar}\right).$$

The n-th wave function has exactly n nodes, consistent with the general principle that higher-energy states oscillate more rapidly.

18 3. Ubiquity of the Harmonic Oscillator

The harmonic oscillator appears throughout physics:

[nosep]

- **Molecular vibrations**: atoms in a molecule oscillate about equilibrium positions.
- **Phonons**: quantized vibrations of a crystal lattice.
- **Photons**: each mode of the electromagnetic field is a harmonic oscillator; the quanta are photons.
- **Any stable equilibrium**: expanding any smooth potential about a minimum gives $V(x) \approx V_0 + 12V''(x_0)(x - x_0)^2$, which is a harmonic oscillator.

19 4. Spin-$\frac{1}{2}$

19.1 4.1 Intrinsic angular momentum

Particles possess an intrinsic angular momentum called **spin**, which has no classical analog of a spinning object. For an electron (spin-$\frac{1}{2}$), the spin operators satisfy:

$$[S_i, S_j] = i\hbar\, \epsilon_{ijk}\, S_k,$$

the same algebra as orbital angular momentum. The spin quantum number is $s = \frac{1}{2}$, giving two states:

$$S_z\uparrow = +\frac{\hbar}{2}\uparrow, \qquad S_z\downarrow = -\frac{\hbar}{2}\downarrow.$$

19.2 4.2 Pauli matrices

The spin operators for spin-$\frac{1}{2}$ are $S_i = \frac{\hbar}{2}\sigma_i$, where the **Pauli matrices** are:

$$\sigma_x = (\,0\,)\,110, \qquad \sigma_y = (\,0\,) - ii0, \qquad \sigma_z = (\,1\,)\,00 - 1.$$

Key properties:

[nosep]

- $\sigma_i^2 = I$ for each i.
- $\sigma_i\sigma_j + \sigma_j\sigma_i = 2\delta_{ij}I$ (anti-commutation).
- $\sigma_i\sigma_j - \sigma_j\sigma_i = 2i\epsilon_{ijk}\sigma_k$ (commutation).

20 5. Bosons and Fermions — Preview

Particles come in two fundamental types:

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- **Bosons**: integer spin ($s = 0, 1, 2, \ldots$). Examples: photons, phonons, mesons.
- **Fermions**: half-integer spin ($s = \frac{1}{2}, \frac{3}{2}, \ldots$). Examples: electrons, protons, neutrons.

The distinction is *not* merely a label—it has profound consequences for the behavior of identical particles, which will be the subject of the next lecture.

The spin-statistics connection—that integer-spin particles are bosons and half-integer-spin particles are fermions—is one of the deepest results in theoretical physics. Its consequences include the Pauli exclusion principle and the structure of the periodic table.

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Lecture 5: Identical Particles and the Exclusion Principle

Pauli Exclusion, Spin-Statistics, and Atomic Shell Structure Based on Leonard Susskind's *Advanced Quantum Mechanics*

21 1. Identical Particles in Quantum Mechanics

In classical mechanics, identical particles can always be distinguished by tracking their trajectories. In quantum mechanics, this is impossible: there is no trajectory, and identical particles are truly *indistinguishable*.

Definition 5 (Identical particles) *Two particles are* ***identical*** *if no measurement can distinguish one from the other. The physics must be invariant under interchange of the two particles.*

21.1 1.1 Exchange symmetry

Consider a two-particle state $\psi(x_1, x_2)$. Exchanging particles 1 and 2:

$$P_{12}\,\psi(x_1, x_2) = \psi(x_2, x_1).$$

Since $P_{12}^2 = I$, the eigenvalues of P_{12} are ± 1. There are exactly two possibilities:

[nosep]

- **Symmetric**: $\psi(x_2, x_1) = +\psi(x_1, x_2)$ (bosons).
- **Antisymmetric**: $\psi(x_2, x_1) = -\psi(x_1, x_2)$ (fermions).

22 2. The Spin-Statistics Connection

One of the deepest results in physics:

$$Integerspin \iff bosons(symmetric), \qquad Half-integerspin \iff fermions(antisymmetric).$$

This is not an assumption—it is a theorem in relativistic quantum field theory. Key observation: a 2π rotation of a half-integer spin particle introduces a phase factor of -1, which is directly connected to the antisymmetry of the wave function.

23 3. The Pauli Exclusion Principle

Definition 6 (Pauli exclusion principle) *No two identical fermions can occupy the same quantum state.*

Proof from antisymmetry. If both particles are in state ϕ:

$$\psi(x_1, x_2) = \phi(x_1)\phi(x_2).$$

This is symmetric under exchange, so it cannot describe fermions. The antisymmetrized version:

$$\psi_A(x_1, x_2) = \frac{1}{\sqrt{2}}\big[\phi(x_1)\phi(x_2) - \phi(x_2)\phi(x_1)\big] = 0.$$

The state vanishes identically. Two fermions *cannot* be in the same state.

23.1 3.1 Bosons: the opposite behavior

Bosons have no such restriction. In fact, bosons *prefer* to congregate: the probability of finding two bosons in the same state is enhanced relative to distinguishable particles. This is the origin of Bose–Einstein condensation, laser coherence, and the integer quantum Hall effect.

24 4. Multi-Particle States

24.1 4.1 Two distinguishable particles

If particle 1 can be in states $\{\phi_i\}$ and particle 2 in states $\{\chi_j\}$, a general two-particle state is:

$$\psi(x_1, x_2) = \sum_{i,j} c_{ij}\, \phi_i(x_1)\, \chi_j(x_2).$$

24.2 4.2 Two identical bosons

The state must be symmetrized:

$$\psi_S(x_1, x_2) = \frac{1}{\sqrt{2}} \big[\phi_a(x_1)\phi_b(x_2) + \phi_a(x_2)\phi_b(x_1)\big].$$

24.3 4.3 Two identical fermions

The state must be antisymmetrized:

$$\psi_A(x_1, x_2) = \frac{1}{\sqrt{2}} \big[\phi_a(x_1)\phi_b(x_2) - \phi_a(x_2)\phi_b(x_1)\big].$$

This can be written as a **Slater determinant**:

$$\psi_A(x_1, x_2) = \frac{1}{\sqrt{2}} \phi_a(x_1)\phi_b(x_1)\phi_a(x_2)\phi_b(x_2).$$

The determinant automatically vanishes if $a = b$, enforcing the exclusion principle.

25 5. Atomic Shell Structure

25.1 5.1 Filling order

In a hydrogen-like atom, each electron state is labeled by quantum numbers (n, ℓ, m, m_s). The Pauli exclusion principle dictates that each combination of quantum numbers can be occupied by at most one electron.

For a given n and ℓ:

[nosep]

- There are $2\ell + 1$ values of m.

- Each m has 2 spin states ($m_s = \pm\frac{1}{2}$).
- Total: $2(2\ell + 1)$ electrons per subshell.

25.2 5.2 Shell capacities

Subshell	ℓ	**States**	**Max electrons**
s	0	1	2
p	1	3	6
d	2	5	10
f	3	7	14

25.3 5.3 The periodic table

The structure of the periodic table arises directly from the Pauli exclusion principle applied to atomic electrons. As atomic number Z increases, electrons fill successively higher energy levels. The chemical properties of an element are determined primarily by the number of electrons in the outermost (valence) shell:

[nosep]

- Noble gases: filled shells, chemically inert.
- Alkali metals: one electron beyond a filled shell, highly reactive.
- Halogens: one electron short of a filled shell, highly reactive.

The Pauli exclusion principle, combined with the shell structure of atoms, explains the periodicity of chemical properties—one of the great triumphs of quantum mechanics. Next, we move to the formalism of second quantization, which provides a natural language for many-particle systems.

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Lecture 6: Second Quantization

From Harmonic Oscillators to Quantum Fields Based on Leonard Susskind's *Advanced Quantum Mechanics*

26 1. Review of the Harmonic Oscillator Algebra

The algebraic structure of a single harmonic oscillator:

$$[a, a^\dagger] = 1, \qquad N = a^\dagger a, \qquad H = \hbar\omega\,(N + 12)\,.$$

The states $0, 1, 2, \ldots$ are the occupation-number eigenstates, with $a^\dagger n = \sqrt{n+1}\,n + 1$ and $an = \sqrt{n}\,n - 1$.

27 2. Many Independent Oscillators

Now consider a collection of independent harmonic oscillators, each labeled by an index i and having its own frequency ω_i. Each oscillator has its own creation and annihilation operators:

$$[a_i, a_j^\dagger] = \delta_{ij}, \qquad [a_i, a_j] = 0, \qquad [a_i^\dagger, a_j^\dagger] = 0.$$

The total Hamiltonian is:

$$H = \sum_i \hbar\omega_i \left(a_i^\dagger a_i + \frac{1}{2} \right).$$

A general state is specified by the occupation numbers of each oscillator:

$$n_1, n_2, n_3, \ldots = \frac{(a_1^\dagger)^{n_1}}{\sqrt{n_1!}} \frac{(a_2^\dagger)^{n_2}}{\sqrt{n_2!}} \cdots 0, 0, 0, \ldots.$$

This is the **Fock space** representation.

28 3. Reinterpretation: Particles as Excitations

The key insight of second quantization is a *change of language*:

Oscillator language	**Particle language**
Oscillator i in n-th excited state	n particles in single-particle state i
$a_i^\dagger$ raises excitation of oscillator i	$a_i^\dagger$ creates a particle in state i
a_i lowers excitation of oscillator i	a_i annihilates a particle in state i
$N_i = a_i^\dagger a_i$ counts excitation level	N_i counts particles in state i
Vacuum 0 (all oscillators in ground state)	Empty space (no particles)

The mathematics is identical; only the interpretation changes. The occupation number n_i is now the number of particles in single-particle state i.

28.1 3.1 Total particle number

The total number of particles is:

$$\hat{N}_{total} = \sum_i a_i^\dagger a_i.$$

This operator counts the sum of all occupation numbers.

29 4. Bosonic Many-Particle States

For bosons, the commutation relations $[a_i, a_j^\dagger] = \delta_{ij}$ allow any number of particles in each state. The symmetry of the wave function under particle exchange is automatically guaranteed by the commutation of creation operators:

$$a_i^\dagger a_j^\dagger = a_j^\dagger a_i^\dagger.$$

Creating particle i then particle j is the same as creating j then i—the state is automatically symmetric.

30 5. Creation and Annihilation Operators for Particles

Definition 7 (Creation operator) *The operator $a_i^\dagger$* ***creates*** *a particle in single-particle state i:*

$$a_i^\dagger n_1, \ldots, n_i, \ldots = \sqrt{n_i + 1}\, n_1, \ldots, n_i + 1, \ldots.$$

Definition 8 (Annihilation operator) *The operator a_i* ***annihilates*** *(removes) a particle from single-particle state i:*

$$a_i n_1, \ldots, n_i, \ldots = \sqrt{n_i}\, n_1, \ldots, n_i - 1, \ldots.$$

If there is no particle in state i ($n_i = 0$), then a_i gives zero.

31 6. Why "Second Quantization"?

The term "second quantization" is somewhat misleading. There is really only one quantization—the standard rules of quantum mechanics. The name arose historically because the procedure *looks like* quantizing the already-quantized single-particle wave function by promoting it to an operator.

The real power of the formalism is:

[nosep]

1. It automatically handles identical-particle symmetry (symmetric for bosons, antisymmetric for fermions).

2. It naturally describes systems with a *variable* number of particles (creation and annihilation).

3. It provides the natural language for quantum field theory.

In the next lecture, we will construct the quantum field operator $\Psi(x)$ as a superposition of annihilation operators weighted by single-particle wave functions, completing the bridge from particles to fields.

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Lecture 7: Quantum Fields

Field Operators, Particle Density, and the Many-Body Hamiltonian

Based on Leonard Susskind's *Advanced Quantum Mechanics*

32 1. Motivation: Variable Particle Number

Many physical processes involve changes in the number of particles:

[nosep]

- An atom emits a photon (one photon is created).
- A neutron decays into a proton, electron, and neutrino.
- Particle–antiparticle pair creation and annihilation.

The single-particle Schrödinger equation cannot describe such processes. We need a framework in which the number of particles is a dynamical variable. This is precisely what the field-operator formalism provides.

33 2. The Field Operator

Definition 9 (Field operator) *The **annihilation field operator** at position x is:*

$$\Psi(x) = \sum_i a_i \, \psi_i(x),$$

*where $\{a_i\}$ are the annihilation operators for single-particle states $\{i\}$, and $\{\psi_i(x)\}$ are the corresponding single-particle wave functions. The **creation field operator** is its Hermitian conjugate:*

$$\Psi^\dagger(x) = \sum_i a_i^\dagger \, \psi_i^*(x).$$

33.1 2.1 What the field operators do

[nosep]

- $\Psi^\dagger(x)0$ creates a particle at position x from the vacuum.
- $\Psi(x)$ annihilates a particle at position x.
- These are *operators*, not wave functions. They act on the Fock space of many-particle states.

33.2 2.2 Contrast with single-particle wave functions

Wave function $\psi(x)$	**Field operator** $\Psi(x)$
Complex number at each x	Operator at each x
Describes one particle	Describes any number of particles
$\lvert\psi(x)\rvert^2$ = probability density	$\Psi^\dagger(x)\Psi(x)$ = particle density operator
Not an observable	Built from observables

34 3. Particle Density and Number Operator

34.1 3.1 Particle density operator

The operator

$$\hat{n}(x) = \Psi^\dagger(x)\Psi(x)$$

is the **particle number density** at position x. Its expectation value $\hat{n}(x)$ gives the average number of particles per unit length (or volume) at x.

34.2 3.2 Total number operator

Integrating the density over all space gives the total particle number:

$$\hat{N} = \int dx\ \Psi^\dagger(x)\Psi(x) = \sum_i a_i^\dagger a_i.$$

This is the sum of the number operators for each single-particle state, confirming the consistency of the formalism.

35 4. The Many-Body Hamiltonian

35.1 4.1 Kinetic energy

The kinetic energy of a system of non-interacting particles is:

$$H_{kin} = \int dx\ \Psi^\dagger(x)\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\right)\Psi(x).$$

This operator sums the kinetic energy $p^2/(2m)$ of each particle.

35.2 4.2 External potential

If the particles move in an external potential $V(x)$:

$$H_V = \int dx\, V(x)\, \Psi^\dagger(x)\Psi(x).$$

This simply counts the particles at each point x and assigns each one the potential energy $V(x)$.

35.3 4.3 Full single-particle Hamiltonian

Combining kinetic and potential terms:

$$H = \int dx\, \Psi^\dagger(x) \left[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x) \right] \Psi(x).$$

This is the second-quantized Hamiltonian for non-interacting particles in a potential. The expression in brackets is just the single-particle Hamiltonian acting between the creation and annihilation field operators.

36 5. Physical Interpretation

The second-quantized Hamiltonian encodes all single-particle physics:

[nosep]

1. The field operator $\Psi^\dagger(x)$ creates a particle at x.
2. The differential operator $-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)$ acts on that particle.
3. The field operator $\Psi(x)$ closes the expression by annihilating a particle at x.
4. The integral over x sums contributions from all positions.

The result is an operator that, when acting on a many-body state, returns the total energy (kinetic plus potential) of all the particles.

36.1 5.1 Constant potential and rest energy

If $V(x) = mc^2$ (a constant), then $H_V = mc^2\hat{N}$—each particle contributes its rest energy. This term does not exert any force but keeps track of the total rest energy proportional to the number of particles.

In the next lecture, we will explore the physical consequences of this formalism: tunneling, mixing of states, neutrino oscillations, and the properties of the electron. We will also examine how the commutation relations of field operators encode locality and causality.

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Lecture 8: Tunneling, Mixing, and Symmetry

Neutrino Oscillations, the Electron Dipole Moment, and Field Commutators Based on Leonard Susskind's *Advanced Quantum Mechanics*

37 1. Quantum Tunneling and Mixing

37.1 1.1 The double-well potential

Consider a symmetric double-well potential with a barrier between two minima. If the barrier were infinite, there would be two degenerate ground states—one localized on the left (S_L) and one on the right (S_R), each with the same energy E_1.

When the barrier is finite, the wave function can leak through (tunnel). The true energy eigenstates are *not* S_L and S_R but their symmetric and antisymmetric combinations:

$$S_+ = \frac{1}{\sqrt{2}}\big(S_L + S_R\big), \qquad S_- = \frac{1}{\sqrt{2}}\big(S_L - S_R\big),$$

with energies:

$$E_+ = E_1 - \epsilon, \qquad E_- = E_1 + \epsilon,$$

where ϵ is an exponentially small splitting determined by the barrier height and width. The symmetric state S_+ has the lower energy.

37.2 1.2 Oscillations between left and right

If a particle is prepared on the left at $t = 0$:

$$\psi(0) = S_L = \frac{1}{\sqrt{2}}\big(S_+ + S_-\big).$$

Time evolution gives:

$$\psi(t) = \frac{1}{\sqrt{2}}\big(e^{-iE_+t/\hbar}S_+ + e^{-iE_-t/\hbar}S_-\big).$$

After a time $T = \pi\hbar/(2\epsilon)$, the relative phase reverses and the particle is entirely on the right:

$$\psi(T) \propto S_R.$$

The particle oscillates back and forth with period $\pi\hbar/\epsilon$.

38 2. Neutrino Oscillations

Neutrino oscillations are a direct physical realization of the mixing phenomenon:

[nosep]

- The **flavor eigenstates** (electron neutrino ν_e, muon neutrino ν_μ) are the analogs of S_L and S_R.
- The **mass eigenstates** (definite energy/mass) are the analogs of S_+ and S_-.
- The mass eigenstates have slightly different masses, producing a splitting ϵ.

A neutrino created as ν_e (e.g., in a solar nuclear reaction) will oscillate into ν_μ over a characteristic length scale set by ϵ. This was experimentally confirmed and explains the observed deficit of solar electron neutrinos.

38.1 2.1 Spin precession analogy

The same mathematics describes spin precession. A spin-$\frac{1}{2}$ particle in a magnetic field $B = B\hat{x}$:

[nosep]

- The energy eigenstates are $left$ and $right$ (aligned/anti-aligned with $\hat{x}$).
- An electron prepared in $\uparrow$ (z-direction) oscillates between $\uparrow$ and $\downarrow$ with a period set by the magnetic field strength.

39 3. The Electron Dipole Moment and Symmetry

39.1 3.1 Is the electron a sphere?

A common misconception: the absence of an electric dipole moment does not imply spherical symmetry. An object can lack a dipole moment but have higher multipole moments (quadrupole, octupole, etc.). The electron, having spin $\frac{1}{2}$, is intrinsically *not* spherically symmetric—it has a spin axis.

39.2 3.2 Quantum mechanical "sphericity"

For any bosonic system in a state of zero angular momentum ($\ell = 0$), the wave function is rotationally symmetric. Even a dumbbell-shaped molecule in its ground state has a spherically symmetric probability distribution, because $\ell = 0$ implies invariance under all rotations.

Whether this counts as "spherical" depends on the energy resolution of your probe. If the energy gap to the first rotational excitation is much larger than the probe energy, the system appears perfectly spherical.

39.3 3.3 Parity and the electric dipole moment

If the laws of physics are invariant under spatial reflection (parity P), and if there is only one species of electron, then the electron cannot have an electric dipole moment. The argument:

[nosep]

1. Under reflection, the magnetic moment (an axial vector from circulating current) does *not* flip.
2. Under reflection, the electric dipole moment (a polar vector from charge displacement) *does* flip.
3. If the mirror image must be the same particle, the dipole moment must be zero.

However, parity is violated in the weak interaction, so this argument alone is insufficient.

39.4 3.4 Time-reversal symmetry

A stronger constraint comes from time-reversal symmetry T:

[nosep]

1. Under T, currents reverse $\Rightarrow$ the magnetic moment flips.
2. Under T, static charges are unchanged $\Rightarrow$ the electric dipole stays.
3. Again, the mirror image differs from the original unless the dipole is zero.

Time-reversal violation exists in the Standard Model but is extremely small. The predicted electron electric dipole moment is far below current experimental sensitivity. The non-observation of an electric dipole moment constrains extensions of the Standard Model.

40 4. Fourier Transforms and Momentum-Space Fields

40.1 4.1 Single-particle Fourier transform

The position-space and momentum-space wave functions are Fourier conjugates:

$$\tilde{\psi}(p) = \frac{1}{\sqrt{2\pi\hbar}} \int dx\, e^{-ipx/\hbar}\, \psi(x), \qquad \psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int dp\, e^{ipx/\hbar}\, \tilde{\psi}(p).$$

40.2 4.2 Field operators in momentum space

The same relation holds for field operators:

$$\Psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int dp\, \tilde{a}(p)\, e^{ipx/\hbar},$$

where $\tilde{a}(p)$ annihilates a particle of momentum p. The creation field operator is:

$$\Psi^\dagger(x) = \frac{1}{\sqrt{2\pi\hbar}} \int dp\, \tilde{a}^\dagger(p)\, e^{-ipx/\hbar}.$$

The relationship between position-space and momentum-space creation/annihilation operators mirrors the Fourier transform of wave functions.

41 5. Commutation Relations and Locality

41.1 5.1 Field commutators

The commutation relations of the field operators are:

$$[\Psi(x), \Psi^\dagger(y)] = \delta(x-y), \qquad [\Psi(x), \Psi(y)] = 0, \qquad [\Psi^\dagger(x), \Psi^\dagger(y)] = 0.$$

41.2 5.2 Physical meaning: locality

The vanishing of $[\Psi(x), \Psi^\dagger(y)]$ for $x \neq y$ (which holds when considering the real and imaginary parts as observables) has profound physical meaning:

[nosep]

- Measuring the field at point x does not disturb the field at point $y \neq x$.
- There is no Heisenberg-type interference between measurements at different spatial points.
- This is the quantum-mechanical expression of **locality**: operations at one point do not instantaneously affect distant points.

In relativistic quantum field theory, this locality condition extends to spacelike separations, ensuring consistency with special relativity: no signal can propagate faster than the speed of light.

41.3 5.3 Entanglement and locality

Entanglement does *not* violate locality. Measuring an entangled particle at one location does not "kick" the distant particle. The mathematical reason is precisely that field operators at different points commute: $[\Psi(x), \Psi^\dagger(y)] = 0$ for $x \neq y$. Entanglement creates correlations but does not transmit information.

This lecture bridges the abstract formalism of quantum fields to concrete physical phenomena—from neutrino oscillations to fundamental symmetries. Next, we

examine momentum conservation in quantum field theory and the interaction terms that lead to scattering and particle decay.

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Lecture 9: Interactions and the Dirac Equation

Momentum Conservation, Feynman Diagrams, and Relativistic Electrons Based on Leonard Susskind's *Advanced Quantum Mechanics*

42 1. Momentum Conservation in Quantum Field Theory

42.1 1.1 The Hamiltonian in momentum space

Consider the second-quantized Hamiltonian for non-interacting particles with a constant potential mc^2:

$$H = \int dx\, \Psi^\dagger(x) \left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + mc^2 \right] \Psi(x).$$

Expanding the field operators in the momentum basis via Fourier transform:

$$\Psi(x) = \int \frac{dp}{\sqrt{2\pi\hbar}}\, \tilde{a}(p)\, e^{ipx/\hbar}, \qquad \Psi^\dagger(x) = \int \frac{dq}{\sqrt{2\pi\hbar}}\, \tilde{a}^\dagger(q)\, e^{-iqx/\hbar}.$$

42.2 1.2 The delta function and momentum conservation

Substituting and integrating over x:

$$\int dx\, e^{i(p-q)x/\hbar} = 2\pi\hbar\, \delta(p-q).$$

The delta function enforces $p = q$: the momentum of the annihilated particle must equal the momentum of the created particle. The result:

$$H_{mass} = mc^2 \int dp\, \tilde{a}^\dagger(p)\, \tilde{a}(p),$$

which simply counts each particle and assigns it an energy mc^2. The kinetic term similarly gives:

$$H_{kin} = \int dp\, \frac{p^2}{2m}\, \tilde{a}^\dagger(p)\, \tilde{a}(p).$$

42.3 1.3 General momentum conservation rule

For any Hamiltonian of the form $H = \int dx\ \mathcal{H}[\Psi^\dagger(x), \Psi(x)]$—where the integrand involves products of field operators all evaluated at the *same* point x and integrated over all space—the integration over x produces a delta function:

$$\delta\left(\sum_{created} q_i - \sum_{annihilated} p_j\right).$$

This enforces **total momentum conservation**: the sum of momenta of particles created must equal the sum of momenta of particles annihilated. Individual momenta may change, but the total is preserved.

43 2. Interaction Terms

43.1 2.1 Contact scattering

Consider two species of particles (e.g., "electrons" and "protons"). A short-range interaction at the same point:

$$H_{int} = g \int dx\ \Psi_e^\dagger(x)\, \Psi_p^\dagger(x)\, \Psi_e(x)\, \Psi_p(x).$$

This operator:

[nosep]

1. Finds an electron and a proton at the same position x.
2. Annihilates both.
3. Creates an electron and a proton at the same point (possibly with different momenta).

The coupling constant g determines the strength of the interaction. Total momentum is conserved, but individual momenta can change—this is **scattering**.

43.2 2.2 Particle decay

If particle A can decay into particles B and C:

$$H_{decay} = g \int dx\ \left[\Psi_B^\dagger(x)\, \Psi_C^\dagger(x)\, \Psi_A(x) + \Psi_A^\dagger(x)\, \Psi_C(x)\, \Psi_B(x)\right].$$

The first term annihilates an A and creates a B and C. The second term (the Hermitian conjugate) describes the reverse process: $B + C \to A$. The Hamiltonian must be Hermitian, so both directions are always present.

43.3 2.3 Coupling constants

The parameter g is the **coupling constant**:

[nosep]

- Small g: weak interaction, low scattering probability.
- Large g: strong interaction, high scattering probability.
- g is typically determined experimentally.

44 3. Feynman Diagrams

44.1 3.1 Iterating the Hamiltonian

The time-evolution operator expanded to second order:

$$U(t) \approx 1 - i\epsilon H - \frac{\epsilon^2}{2} H^2 + \cdots$$

The first-order term H gives direct processes (single vertex). The second-order term H^2 gives compound processes (two vertices), leading to a rich set of scattering and exchange phenomena.

44.2 3.2 Exchange diagrams

If the Hamiltonian contains a term $A \to B + C$, then H^2 contains the process:

$$A + B \longrightarrow C \longrightarrow A + B,$$

where C is a temporary (virtual) intermediate particle. This is an **exchange diagram**: the interaction between A and B is mediated by the exchange of C. The amplitude carries a factor g^2.

44.3 3.3 Self-energy corrections

Another second-order process: a particle A emits a B and C, which then recombine into A:

$$A \longrightarrow B + C \longrightarrow A.$$

This modifies the effective properties (mass, charge) of particle A. Example: an electron emitting and reabsorbing a virtual photon—the electron has approximately a 1% probability of containing a virtual photon at any given time.

44.4 3.4 Vertices and diagrams

Each vertex in a Feynman diagram corresponds to a term in the Hamiltonian and carries one factor of the coupling constant g. A diagram with n vertices contributes at order g^n. The full amplitude for any process is the sum over all possible diagrams, providing a systematic perturbative expansion.

45 4. Introduction to the Dirac Equation

45.1 4.1 Motivation

For a relativistic particle, the energy–momentum relation is:

$$E^2 = p^2c^2 + m^2c^4.$$

Dirac wanted a wave equation that is *first order* in time (like the Schrödinger equation) rather than second order (like the Klein–Gordon equation $-\hbar^2\partial_t^2\psi = (-\hbar^2c^2\partial_x^2 + m^2c^4)\psi$).

45.2 4.2 The one-dimensional massless case

For a massless particle moving to the right, $E = p$ (setting $c = 1$):

$$i\hbar\frac{\partial\psi}{\partial t} = -i\hbar\frac{\partial\psi}{\partial x}.$$

Solutions are right-moving waves $\psi(x - t)$. Similarly, $E = -p$ gives left-moving waves.

45.3 4.3 Two-component structure

To describe both left- and right-movers, introduce a two-component object $\Psi = (\,\psi\,)_1$
ψ_2 and a matrix α:

$$H = \alpha\, p, \qquad \alpha = (\,1\,)\,00 - 1 = \sigma_z.$$

The eigenvalue $+1$ corresponds to right-movers and -1 to left-movers.

45.4 4.4 Adding mass: the β matrix

To obtain $H^2 = p^2 + m^2$, add a mass term:

$$H = \alpha\, p + \beta\, m.$$

Squaring: $H^2 = \alpha^2p^2 + \beta^2m^2 + (\alpha\beta + \beta\alpha)pm$. We need: $\alpha^2 = 1$, $\beta^2 = 1$, $\alpha\beta + \beta\alpha = 0$. The solution uses two anti-commuting Pauli matrices:

$$\alpha = \sigma_z = (\,1\,)\,00 - 1, \qquad \beta = \sigma_x = (\,0\,)\,110.$$

45.5 4.5 The one-dimensional Dirac equation

The full equation:

$$i\hbar\frac{\partial}{\partial t}\,(\,\psi\,)_1\,\psi_2 = \left[\alpha\left(-i\hbar\frac{\partial}{\partial x}\right) + \beta\, m\right](\,\psi\,)_1\,\psi_2.$$

Written out: $\mathrm{i}\hbar\,\dot{\psi}_1 = -i\hbar\,\psi_1' + m\,\psi_2$,
$i\hbar\,\dot{\psi}_2 = +i\hbar\,\psi_2' + m\,\psi_1$. The mass term βm *couples* the two components: it mixes left-movers and right-movers, giving the particle a mass and allowing it to move slower than the speed of light.

The one-dimensional Dirac equation captures the essential physics: the necessity of a multi-component wave function, the role of the mass as a coupling between chiralities, and the emergence of both positive and negative energy solutions. In the next lecture, we extend this to three dimensions and confront the Dirac sea.

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Lecture 10: The Dirac Equation in Three Dimensions

Fermion Fields, Chirality, Zitterbewegung, and the Dirac Sea Based on Leonard Susskind's *Advanced Quantum Mechanics*

46 1. Fermion Anti-Commutation Relations

46.1 1.1 From bosons to fermions

For bosons, creation and annihilation operators *commute*:

$$[a_i, a_j^\dagger] = \delta_{ij}, \qquad [a_i, a_j] = 0, \qquad [a_i^\dagger, a_j^\dagger] = 0.$$

For fermions, replace commutators with **anti-commutators**:

$$\{a_i, a_j^\dagger\} \equiv a_i a_j^\dagger + a_j^\dagger a_i = \delta_{ij}, \qquad \{a_i, a_j\} = 0, \qquad \{a_i^\dagger, a_j^\dagger\} = 0.$$

46.2 1.2 Pauli exclusion from anti-commutation

From $\{a_i^\dagger, a_i^\dagger\} = 0$, we get:

$$(a_i^\dagger)^2 = 0.$$

Trying to create two fermions in the same state gives zero—the **Pauli exclusion principle** is built into the algebra.

46.3 1.3 Symmetry of creation and annihilation

For bosons, creation and annihilation operators are fundamentally different: the spectrum $n = 0, 1, 2, \ldots$ is unbounded above but bounded below. For fermions, the occupation number is either 0 or 1. This creates a *symmetry* between creation and annihilation: $a^\dagger$ takes $0 \to 1$ and annihilates 1; a takes $1 \to 0$ and annihilates 0. The algebra alone does not distinguish which is which.

46.4 1.4 Fermion field operators

The fermion field operators satisfy anti-commutation relations:

$$\{\Psi(x), \Psi^\dagger(y)\} = \delta(x-y), \qquad \{\Psi(x), \Psi(y)\} = 0, \qquad \{\Psi^\dagger(x), \Psi^\dagger(y)\} = 0.$$

Two-particle states are automatically *antisymmetric*:

$$\Psi^\dagger(x)\Psi^\dagger(y) = -\Psi^\dagger(y)\Psi^\dagger(x).$$

47 2. The Dirac Equation in Three Dimensions

47.1 2.1 The massless case: $H = *\sigma \cdot p$

In three dimensions, the momentum $p = (p_x, p_y, p_z)$ is a vector. For a massless particle, Dirac wrote:

$$H = *\alpha \cdot p = \alpha_x p_x + \alpha_y p_y + \alpha_z p_z.$$

Requiring $H^2 = p^2 = p_x^2 + p_y^2 + p_z^2$ demands: $\alpha_i^2 = I \quad fori = x, y, z$, $\alpha_i\alpha_j + \alpha_j\alpha_i = 0 \quad fori \neq j$. The three Pauli matrices satisfy these relations, so $\alpha_i = \sigma_i$ works:

$$H = *\sigma \cdot p.$$

This is a 2×2 matrix equation acting on a two-component spinor.

47.2 2.2 Helicity

The energy eigenvalues satisfy $E = \pm|p|$. For positive energy ($E > 0$), the spin must be aligned with the momentum ($*\sigma \cdot \hat{p} = +1$). For negative energy ($E < 0$), the spin is anti-aligned.

The quantity $*\sigma \cdot \hat{p}$ (spin projected along the momentum direction) is called the **helicity**. A massless particle has definite helicity: either right-handed (spin along p) or left-handed (spin opposite to p).

47.3 2.3 Adding mass: 4×4 matrices

To add a mass term $H = *\alpha \cdot p + \beta m$, we need a matrix β that anti-commutes with all three α_i. No 2×2 matrix can anti-commute with all three Pauli matrices.

The solution: go to 4×4 matrices. The Dirac matrices in one standard representation:

$$\alpha_i = (\,\sigma\,)_i\, 00 - \sigma_i, \qquad \beta = (\,0\,)\, II0,$$

where each block is a 2×2 matrix. One can verify:

$$\alpha_i^2 = I_{4\times4}, \qquad \beta^2 = I_{4\times4}, \qquad \{\alpha_i, \beta\} = 0, \qquad \{\alpha_i, \alpha_j\} = 2\delta_{ij} I_{4\times4}.$$

47.4 2.4 The full Dirac equation

The three-dimensional Dirac equation:

$$i\hbar\frac{\partial\Psi}{\partial t} = \big(*\alpha\cdot p + \beta\, mc^2\big)\Psi,$$

where Ψ is a four-component **Dirac spinor**. Squaring the Hamiltonian:

$$H^2 = p^2c^2 + m^2c^4,$$

the correct relativistic energy–momentum relation.

48 3. Chirality

The four components of the Dirac spinor encode two binary degrees of freedom:

[nosep]

1. **Spin**: up or down (the physical angular momentum).
2. **Chirality**: right-handed or left-handed (the relationship between spin and momentum).

For a **massless** particle, chirality is conserved—a right-handed particle remains right-handed. The mass term βm is off-diagonal in the chirality basis, so it *mixes* left-handed and right-handed components. In this sense, **mass is a coupling between chiralities**.

The spin operator is *not* $*\alpha$ but rather:

$$S = \frac{\hbar}{2}(\ ) * \sigma 00 * \sigma,$$

which acts identically on both chirality components.

49 4. The Velocity Operator and Zitterbewegung

The velocity operator for a Dirac particle is obtained from:

$$\dot{x}_i = \frac{i}{\hbar}[H, x_i] = \alpha_i.$$

This is remarkable: the velocity is the matrix α_i, whose eigenvalues are ± 1 (in units of c). The particle always moves at the speed of light instantaneously!

However, α_i does *not* commute with H (because α_x does not commute with α_y or α_z), so the velocity is not conserved. The particle undergoes a rapid oscillatory motion called **Zitterbewegung** ("trembling motion").

The time-averaged velocity is p/m for slow particles, which is the expected classical result. The Zitterbewegung is a high-frequency quantum oscillation superimposed on this smooth classical motion.

50 5. Negative Energy and the Dirac Sea

50.1 5.1 The problem

The Dirac equation has solutions with $E = \pm\sqrt{p^2 + m^2}$. Negative-energy states are problematic: a particle could radiate energy indefinitely, cascading to ever-more-negative energies.

50.2 5.2 Dirac's solution

Dirac proposed that the vacuum is the state in which **all negative-energy states are filled** with fermions (the **Dirac sea**). The Pauli exclusion principle prevents any additional particles from entering these states, stabilizing the vacuum.

This would not work for bosons, since any number of bosons can occupy the same state—there would be no bottom.

50.3 5.3 Positrons as holes

A **hole** in the Dirac sea—a missing negative-energy electron—behaves as a particle with:

[nosep]

- **Positive energy** (removing negative energy increases the total).
- **Positive charge** (a missing negative charge).
- The same mass as the electron.

This is the **positron**—the antiparticle of the electron, predicted by Dirac in 1931 and discovered by Anderson in 1932.

50.4 5.4 Pair creation and annihilation

A photon with sufficient energy ($E \geq 2mc^2$) can excite an electron from the Dirac sea to a positive-energy state, creating an electron–positron pair. The reverse process—an electron falling into a hole—is pair annihilation, producing photons.

51 6. The Modern Viewpoint

Modern quantum field theory dispenses with the Dirac sea picture. Instead:

[nosep]

- The annihilation operator for a negative-energy electron is *relabeled* as a creation operator for a positron.

- The field operator $\Psi(x)$ is split into positive-frequency (electron annihilation) and negative-frequency (positron creation) parts.
- The vacuum is simply the state annihilated by all annihilation operators—no sea required.

The physics is identical; only the language changes.

51.1 6.1 Connection to solid-state physics

The same Dirac-sea picture appears in condensed matter:

[nosep]

- The **Fermi sea**: electrons fill energy levels up to the Fermi energy.
- A missing electron below the Fermi level is a **hole**—a positively charged quasiparticle.
- Excitations above the Fermi level are electrons; excitations below are holes.

Dirac's idea preceded the solid-state application by decades.

51.2 6.2 Vacuum energy

The filled Dirac sea carries an infinite negative energy. Bosonic fields contribute infinite positive zero-point energy ($\frac{1}{2}\hbar\omega$ per mode). In practice, these infinities are subtracted (renormalized). The residual vacuum energy is the **cosmological constant problem**—one of the deepest unsolved problems in physics.

This concludes the Advanced Quantum Mechanics lecture series. Starting from the review of quantum mechanics and symmetries, we have built up through angular momentum, identical particles, and second quantization to the Dirac equation and the prediction of antimatter—one of the greatest achievements of theoretical physics.